

SPRINT 2 - Xavier Palet Bustillo

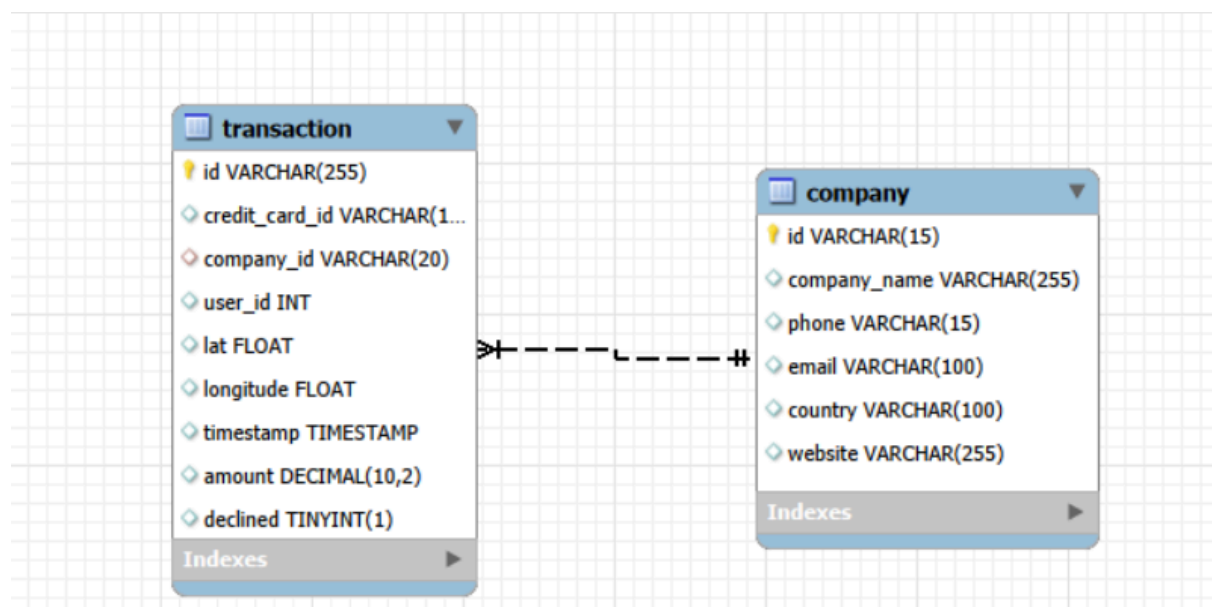
Revisión peer to peer: Gledson Vargas

Nivell 1

Ejercicio 1

En la tabla "company" tenemos una lista con 100 empresas, donde la PRIMARY KEY es el ID de la empresa, con sus respectivos nombres, número de teléfono, email, del país al que pertenecen y sus páginas web.

En la tabla "transaction" tenemos una lista de 49761 filas, donde las PRIMARY KEYS son el ID de la transacción y el ID de la empresa que ha efectuado la transacción (esta id de la empresa se relaciona con la tabla de "company", la tabla también muestra el id de la tarjeta de crédito, el id del usuario que ha efectuado la transacción, las coordenadas de donde se ha efectuado la transacción, la fecha y la hora exacta, la cantidad y si se ha declinado o no la transacción.



La relación es de muchos (de la tabla "transaction") a uno (de la tabla "company").

Ejercicio 2

- Llistat dels països que estan generant vendes:

```

8
9   # Exercici 2
10
11 • SELECT DISTINCT c.country
12   FROM company c
13   JOIN transaction t
14   ON c.id = t.company_id
15   ORDER BY c.country DESC;
16

```

	country
▶	United States
	United Kingdom
	Sweden
	Spain
	Norway
	New Zealand
	Netherlands
	Italy
	Ireland
	Germany

100147 11:42:08 SELECT DISTINCT c.country FROM company c JOIN transaction t ON c.id = t.company_id ORDER BY ... 15 row(s) returned

0.000 sec / 0.000 sec

- Des de quants països es generen les vendes:

```

17 • SELECT COUNT(DISTINCT c.country) AS TOTAL_PAISOS
18     FROM company c
19     JOIN transaction t
20     ON c.id = t.company_id;

```

	TOTAL_PAISOS
▶	15

transaction t ON c.i... 1 row(s) returned 0.031 sec / 0.000 sec

- Identifica la companyia amb la mitjana més gran de vendes:

```

22 • SELECT company_name, avg(amount)
23     FROM transaction t
24     JOIN company c
25     on t.company_id = c.id
26     GROUP BY company_id
27     ORDER BY avg(amount) desc
28     LIMIT 1;

```

	company_name	avg(amount)
▶	Ac Fermentum Incorporated	284.867160

28 11:03:49 SELECT company_name, avg(amount) FROM transaction t JOIN company c on t.company_id = c.id GRO... 1 row(s) returned 0.141 sec / 0.000 sec

Ejercicio 3

- Mostra totes les transaccions realitzades per empreses d'Alemanya:

```

34 • SELECT *
35 FROM transaction
36 WHERE company_id IN (
37 SELECT id
38 FROM company
39 WHERE country = 'Germany'
40 );

```

	id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
▶	00138D3B-206D-4C03-94B7-63A267EB9B4	CcS-4899	b-2222	318	41.3781	12.447	2020-03-25 10:43:43	426.36	0
	0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9	CcS-5070	b-2222	489	41.3814	2.18176	2020-12-17 18:15:37	316.90	0
	00201A11-2E62-44C4-941D-198FC8DB77F0	CcU-3512	b-2222	193	55.5704	-3.65129	2021-01-22 23:44:27	453.04	0
	00235618-0A5C-4D49-9DCB-B3A9405D8923	CcS-8137	b-2222	3556	59.8421	18.729	2020-09-09 15:43:19	263.14	0
	005A5A7B-1F1A-4B6C-9B15-1625A78C9C38	CcS-8998	b-2222	4417	41.1591	-8.63905	2024-05-15 09:10:11	442.01	0
	00687139-48B2-4FFA-8E73-B20376F04AB4	CcS-4870	b-2222	289	51.1966	10.4669	2019-03-09 19:37:49	524.84	0
	0074F4DD-32F1-4827-8758-55896314623A	CcS-8081	b-2222	3500	39.7016	-8.50325	2016-12-26 23:06:57	491.90	0
	00AAB9CD-39D6-4DCB-8A1D-13BE73DC90A9	CcS-6797	b-2222	2216	55.7652	-3.76245	2021-04-25 03:06:59	167.15	0
	00F00014-0000-4700-AB00-20F000000000	CcS-4899	b-2222	489	39.708	0.12003	2019-03-23 15:05:16	141.66	0

100150 11:45:13 SELECT * FROM transaction WHERE company_id IN (SELECT id FROM company WHERE country = 'G... 13291 row(s) returned

0.000 sec / 0.047 sec

- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions:

```

44 • SELECT c.company_name
45 FROM company c
46 WHERE c.id IN (
47 SELECT t.company_id
48 FROM transaction t
49 GROUP BY t.company_id
50 HAVING AVG(t.amount) > (
51 SELECT AVG(amount)
52 FROM transaction
53 )
54 );

```

	company_name
▶	Ac Fermentum Incorporated
	Magna A Neque Industries
	Fusce Corp.
	Ante Iaculis Nec Foundation
	Donec Ltd
	Nascetur Ridiculus Mus Inc.
	Vestibulum Lorem PC

100160 12:07:57 SELECT c.company_name FROM company c WHERE c.id IN (SELECT t.company_id FROM transa... 47 row(s) returned 0.203 sec / 0.000 sec

- Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses:

```

58 • SELECT company_id, company_name
59 FROM company
60 WHERE company_id NOT IN (
61     SELECT business_id
62     FROM transaction
63     WHERE business_id IS NOT NULL
64 );

```

	company_id	company_name
*	NULL	NULL

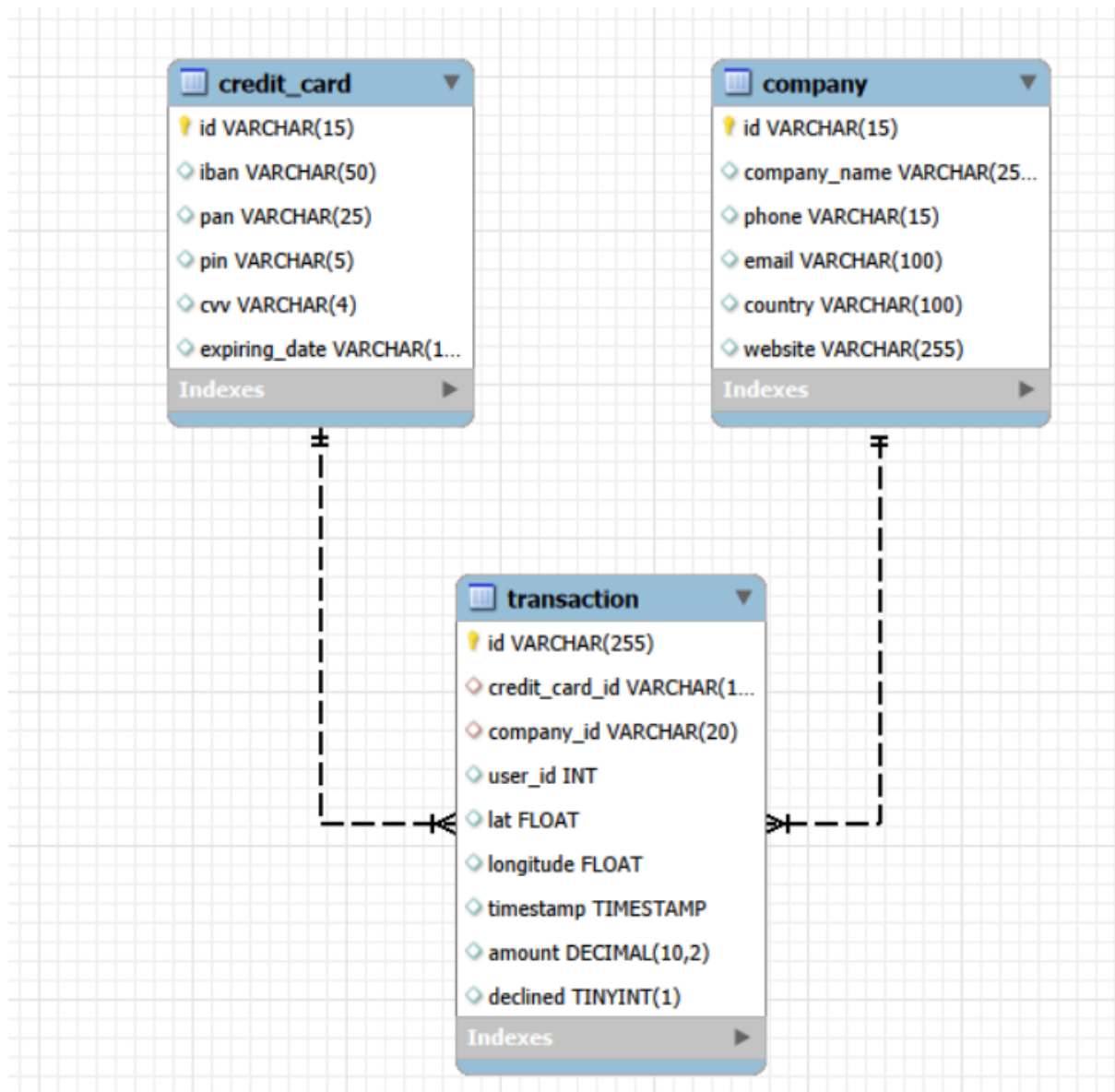
19 11:22:11 SELECT company_id, company_name FROM company WHERE company_id NOT IN (SELECT business_id ... 0 row(s) returned 0.062 sec / 0.000 sec

Ejercicio 4

- La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat

"dades_introduir_credit". Recordar mostrar el diagrama i realitzar una breu descripció d'aquest:

```
72 • CREATE TABLE credit_card (  
73     id VARCHAR(15) PRIMARY KEY,  
74     iban VARCHAR(50),  
75     pan VARCHAR(25),  
76     pin VARCHAR(5),  
77     cvv VARCHAR(5),  
78     expiring_date VARCHAR(20)  
79 );  
80  
81 • ALTER TABLE transaction  
82 ADD CONSTRAINT fk_transaction_credit_card  
83 FOREIGN KEY (credit_card_id) REFERENCES credit_card(id);  
84
```



En este nuevo diagrama tenemos la nueva tabla “credit_card” donde el “ID” es la PRIMARY KEY y se conecta con esta a la tabla central “transaction” a través de la FOREIGN KEY “credit_card_id”, en una relación de 1 a muchos, ya que una tarjeta puede estar asociada a múltiples transacciones.

Ejercicio 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar:

```

87 • SELECT *
88   from credit_card
89   WHERE id = 'CcU-2938';
90
91 • UPDATE credit_card
92   SET iban = 'TR323456312213576817699999'
93   WHERE id = 'CcU-2938';
94

```

	id	iban	pan	pin	cvv	expiring_date
▶	CcU-2938	TR323456312213576817699999	5424465566813633	3257	984	10/30/22
*	NULL	NULL	NULL	NULL	NULL	NULL

105176 13:17:14 SELECT * from credit_card WHERE id = 'CcU-2938' 1 row(s) returned 0.000 sec / 0.000 sec

Ejercicio 6

En la taula "transaction" ingressa una nova transacció amb la següent informació:


```

98 • INSERT INTO company (id, company_name, phone, email, country, website)
99   VALUES (
100     'b-9999',
101     'ghost company',
102     '000000000',
103     'ghost@email.com',
104     'Ghost Country',
105     'www.ghost.com'
106   );

```

```

108 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date)
109   VALUES (
110     'CcU-9999',
111     'ES0000000000000000000000',
112     '0000000000000000',
113     '0000',
114     '000',
115     '12/99'
116   );

```

```

118 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)
119   VALUES (
120     '108B1D1D-5B23-A76C-55EF-C568E49A99DD',
121     'CcU-9999',
122     'b-9999',
123     9999,
124     829.999,
125     -117.999,
126     111.11,
127     0
128   );

```

```

130 • SELECT *
131   FROM transaction
132  WHERE credit_card_id = 'CcU-9999';

```

	id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
▶	108B1D1D-5B23-A76C-55EF-C568E49A99DD	CcU-9999	b-9999	9999	829.999	-117.999	NULL	111.11	0
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```

105179 13:32:52 INSERT INTO company (id, company_name, phone, email, country, website) VALUES ( 'b-9999', 'g... 1 row(s) affected 0.032 sec
105180 13:32:55 INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ( 'CcU-9999', 'ES0000000... 1 row(s) affected 0.000 sec
105181 13:32:57 INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VAL... 1 row(s) affected 0.016 sec
105182 13:34:08 SELECT * FROM transaction WHERE credit_card_id = 'CcU-9999' 1 row(s) returned 0.000 sec / 0.000 sec

```

Ejercicio 7

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recordra mostrar el canvi realitzat:

```
137 • ALTER TABLE credit_card
138     DROP COLUMN pan;
139
140 • SELECT *
141     FROM credit_card;
```

	id	iban	pin	cvv	expiring_date
▶	CcS-4857	XX4857591835292505850771	1819	467	09/27/25
	CcS-4858	XX8581768137002436094025	3964	817	12/28/28
	CcS-4859	XX7826930491423553609370	4983	277	11/26/26
	CcS-4860	XX5559590368835304645299	6876	661	07/27/27
	CcS-4861	XX2035182877195191627307	5710	398	04/25/26
	CcS-4862	XX4774721462463645409758	4042	174	11/27/26

2 10:11:01 SELECT * FROM credit_card 5001 row(s) returned 0.000 sec / 0.000 sec

Ejercicio 8

Descarrega els arxius CSV que trobaràs a l'apartat de recursos:

american_users.csv
european_users.csv
companies.csv
credit_cards.csv
transactions.csv

Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguem realitzar les següents consultes:
La taula de products.csv l'utilitzarem més endavant.

```
145 • CREATE DATABASE IF NOT EXISTS star_schema_db;
146 • USE star_schema_db;
147
148 • CREATE TABLE user (
149     id INT PRIMARY KEY,
150     name VARCHAR(50),
151     surname VARCHAR(50),
152     phone VARCHAR(50),
153     email VARCHAR(100),
154     birth_date VARCHAR(20),
155     country VARCHAR(50),
156     city VARCHAR(50),
157     postal_code VARCHAR(20),
158     address VARCHAR(100),
159     signup_date VARCHAR(20),
160     user_segment VARCHAR(50),
161     income_band VARCHAR(50),
162     region VARCHAR(50)
163 );
164 • CREATE TABLE company (
165     company_id VARCHAR(15) PRIMARY KEY,
166     company_name VARCHAR(100),
167     phone VARCHAR(50),
168     email VARCHAR(100),
169     country VARCHAR(50),
170     website VARCHAR(100),
171     merchant_category VARCHAR(50),
172     merchant_price_position VARCHAR(50)
173 );
```

```
175 ● ○ CREATE TABLE credit_card (  
176     id VARCHAR(15) PRIMARY KEY,  
177     user_id INT,  
178     iban VARCHAR(50),  
179     pan VARCHAR(25),  
180     pin VARCHAR(5),  
181     cvv VARCHAR(4),  
182     track1 VARCHAR(100),  
183     track2 VARCHAR(100),  
184     expiring_date VARCHAR(10),  
185     card_type VARCHAR(20),  
186     card_renewal_flag INT  
187 );
```

```

189 ● ○ CREATE TABLE transaction (
190     id VARCHAR(50) PRIMARY KEY,
191     card_id VARCHAR(15),
192     business_id VARCHAR(50),
193     user_id INT,
194     timestamp VARCHAR(25),
195     amount DECIMAL(10, 2),
196     declined INT,
197     product_ids VARCHAR(50),
198     lat DECIMAL(10, 6),
199     longitude DECIMAL(10, 6),
200     discount_amount DECIMAL(10, 2),
201     tax_amount DECIMAL(10, 2),
202     shipping_amount DECIMAL(10, 2),
203     channel VARCHAR(20),
204     campaign_id VARCHAR(50),
205     device_type VARCHAR(20),
206     is_international INT,
207     decline_reason VARCHAR(50),
208     distance_km DECIMAL(10, 2),
209     FOREIGN KEY (card_id) REFERENCES credit_card(id),
210     FOREIGN KEY (business_id) REFERENCES company(company_id),
211     FOREIGN KEY (user_id) REFERENCES user(id)
212 );

```

```

226 ● LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv'
227 INTO TABLE user
228 FIELDS TERMINATED BY ','
229 ENCLOSED BY '"'
230 LINES TERMINATED BY '\n'
231 IGNORE 1 ROWS
232 (id, name, surname, phone, email, birth_date, country, city, postal_code, address, signup_date, user_segment, income_band)
233 SET region = 'Europa';

```

```

219 ● LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__companies.csv'
220 INTO TABLE company
221 FIELDS TERMINATED BY ','
222 ENCLOSED BY '"'
223 LINES TERMINATED BY '\n'
224 IGNORE 1 ROWS;

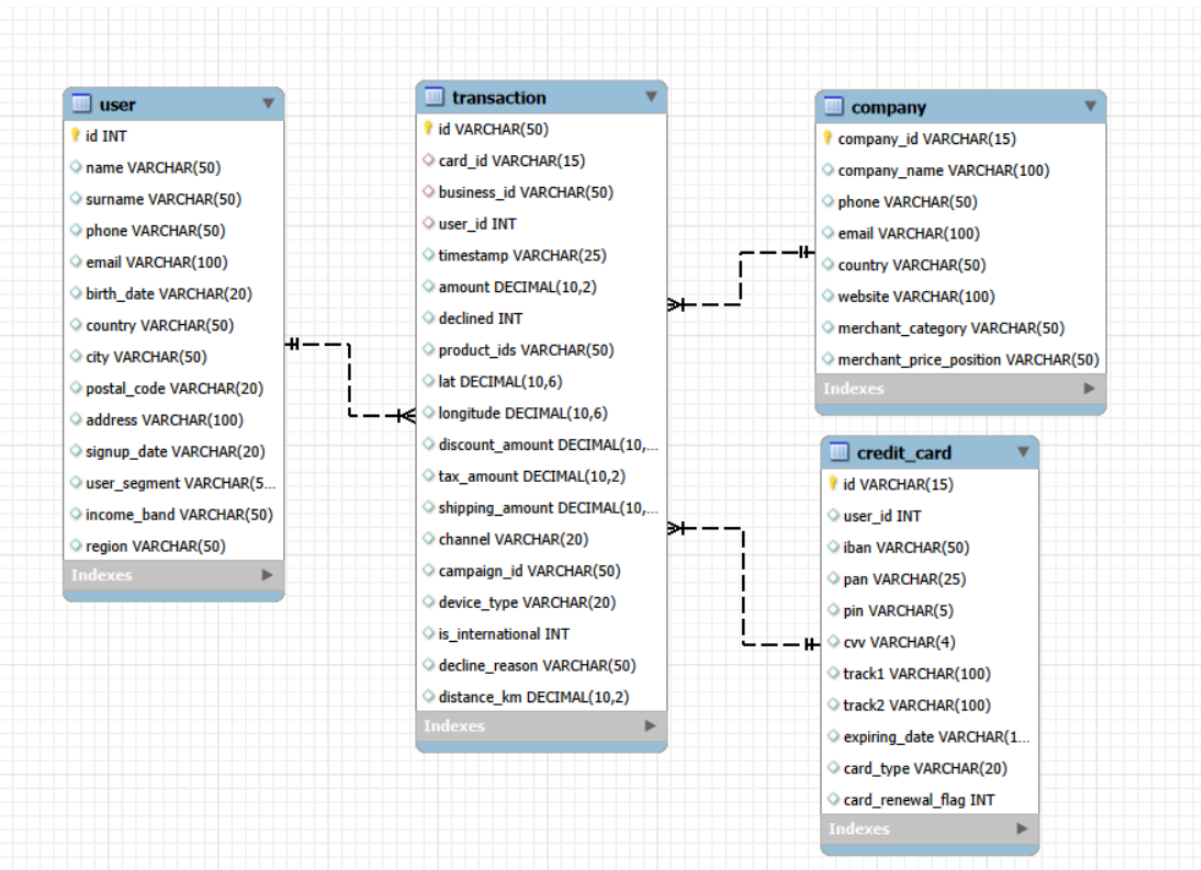
```

```

235 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__american_users.csv'
236 INTO TABLE user
237 FIELDS TERMINATED BY ','
238 ENCLOSED BY '"'
239 LINES TERMINATED BY '\n'
240 IGNORE 1 ROWS
241 (id, name, surname, phone, email, birth_date, country, city, postal_code, address, signup_date, user_segment, income_band)
242 SET region = 'America';

244 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__transactions.csv'
245 INTO TABLE transaction
246 FIELDS TERMINATED BY ';'
247 ENCLOSED BY '"'
248 LINES TERMINATED BY '\n'
249 IGNORE 1 ROWS
250 (id, card_id, business_id, timestamp, amount, declined, product_ids, user_id, lat, longitude, discount_amount, tax_amount, shipping_amount, channel, campaign_id, device_type, is_international, decline_reason, distance_km);

```



Ejercicio 9

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules:

```
223 ● SELECT *
224     FROM user u
225     WHERE u.id IN (
226         SELECT user_id
227         FROM transaction
228         WHERE declined = 0
229         GROUP BY user_id
230         having count(user_id) > 80
231     );
```

id	name	surname	phone	email	birth_date	country	city	postal_code	address	signup_date	user_segment	income_band	region
185	Molly	Gillam	0800 120 8023	donec@outlook.couk	Dec 21, 1993	United Kingdom	London	EC1A 1BB	P.O. Box 202, 5638 M Rd.	2013-08-11	frequent_shopper	high	Europa
289	Dxwgi	Hwczu	+98-309-8797	dxwgi.hwczu@example.com	Aug 20, 1976	Germany	Stuttgart	70173	82 Hwczu Street	2013-12-11	occasional	low	Europa
318	Bnyr	Astuw	+33-120-9644	bnyr.astuw@example.com	May 3, 1974	Italy	Genoa	16100	53 Astuw Street	2013-05-29	frequent_shopper	very_high	Europa
454	Sfzzoh	Xgvfridxs	+58-495-3945	sfzzoh.xgvfridxs@example.com	Aug 28, 1962	Poland	Gdansk	80-001	52 Xgvfridxs Street	2014-08-30	established	medium	Europa

147 13:54:03 SELECT * FROM user u WHERE u.id IN (SELECT user_id FROM transaction WHERE declined = 0 GROUP ... 4 row(s) returned 0.234 sec / 0.000 sec

Ejercicio 10

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules:

```

236 ● SELECT iban, avg(amount)
237 FROM company c
238 JOIN transaction t
239 on c.company_id = t.business_id
240 JOIN credit_card cr
241 on t.card_id = cr.id
242 WHERE company_name = 'Donec Ltd'
243 GROUP BY iban;

```

	iban	avg(amount)
▶	CA901355773586771626108990	361.010000
	US729212642816955651428791	142.560000
	US899450803230933276819251	246.900000
	US581097432668325541142297	154.010000
	CA441226267105857473570574	250.510000
	CA332171962352073015086775	203.950000
	US985359628701198880628720	438.160000

148 13:55:36 SELECT iban, avg(amount) FROM company c JOIN transaction t on c.company_id = t.business_id JOIN credit_c... 371 row(s) returned

0.015 sec / 0.000 sec

Nivell 2

Ejercicio 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes:


```

309 ● SELECT
310     t.*,
311     top_dies.total_vendes
312 FROM
313     transaction t
314 JOIN (
315     SELECT
316         DATE(timestamp) AS dia,
317         SUM(amount) AS total_vendes
318     FROM
319         transaction
320     WHERE declined = 0
321     GROUP BY
322         DATE(timestamp)
323     ORDER BY
324         total_vendes DESC
325     LIMIT 5
326 ) AS top_dies ON DATE(t.timestamp) = top_dies.dia
327 ORDER BY
328     top_dies.total_vendes DESC,
329     t.timestamp ASC;

```

mp	amount	declined	product_ids	lat	longitude	discount_amount	tax_amount	shipping_amount	channel	campaign_id	device_type	is_international	decline_reason	distance_km	total_venc
2-20 00:17:16	724.96	0	29, 33, 79, 27	41.769532	12.161100	5.85	130.73	0.00	store	NO_CAMPAIGN	pos	0		35.54	15728.40
2-20 00:42:06	633.36	0	88, 14, 56, 59	39.943873	-75.156334	5.97	42.81	0.00	online	NO_CAMPAIGN	mobile	0		1.23	15728.40
2-20 01:23:26	401.98	0	100, 5, 61, 79	43.642464	-79.356274	3.73	33.19	0.00	marketplace	NO_CAMPAIGN	tablet	0		2.47	15728.40
2-20 01:45:38	578.44	0	92, 97, 49, 62	51.872043	19.091558	5.71	108.16	0.00	online	NO_CAMPAIGN	mobile	0		6.42	15728.40

#	Time	Action	Message	Duration / Fetch
38	11:01:10	SELECT t.*, top_dies.total_vendes FROM transaction t INNER JOIN (-- Esta es la SUBCONSULTA	219 row(s) returned	0.203 sec / 0.000 sec

Ejercicio 2

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat:

```

324 • SELECT c.company_name, c.phone, c.country, DATE(timestamp) as fecha, t.amount as cantidad
325 from company c
326 JOIN transaction t
327 on c.company_id = t.business_id
328 WHERE DATE(timestamp) IN ('2015-04-29', '2018-07-20', '2024-03-13') and t.amount BETWEEN 350 and 400 and t.declined = 0
329 ORDER BY cantidad desc;

```

	company_name	phone	country	fecha	cantidad
►	Fringilla LLC	+1 150 784 1716	New Zealand	2015-04-29	390.33
	Auctor Mauris Vel LLP	+32 133 447 8239	United States	2024-03-13	383.58
	Mauris Institute	+75 473 619 9472	Sweden	2015-04-29	369.20
	Pede Cum Ltd	+26 625 798 6126	Norway	2018-07-20	358.64
	Netus Et Malesuada Ltd	+52 818 402 1638	Netherlands	2024-03-13	352.21
	Tempor Diam Institute	+89 848 141 3136	Netherlands	2024-03-13	350.71

59 11:56:06 SELECT c.company_name, c.phone, c.country, DATE(timestamp) as fecha, t.amount as cantidad from company ... 6 row(s) returned

0.078 sec / 0.000 sec

Ejercicio 3

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys:

```

336 • SELECT c.company_name, count(t.id) AS total_transacciones,
337 CASE
338     WHEN COUNT(t.id) >= 400 THEN 'Sí (400 o más)'
339     ELSE 'No (Menos de 400)'
340 END AS supera_objetivo
341 FROM transaction t
342 JOIN company c
343 on t.business_id = c.company_id
344 GROUP by business_id;

```

	company_name	total_transacciones	supera_objetivo
►	Ac Fermentum Incorporated	2401	Sí (400 o más)
	Magna A Neque Industries	410	Sí (400 o más)
	Fusce Corp.	447	Sí (400 o más)
	Convallis In Incorporated	1514	Sí (400 o más)
	Ante Iaculis Nec Foundation	472	Sí (400 o más)
	Donec Ltd	449	Sí (400 o más)
	Sed Nunc Ltd	1541	Sí (400 o más)
	Amet Nulla Donec Corporation	1511	Sí (400 o más)
	Nascetur Ridiculus Mus Inc.	424	Sí (400 o más)
	Vestibulum Lorem PC	450	Sí (400 o más)

65 12:46:29 SELECT c.company_name, count(t.id) AS total_transacciones, CASE WHEN COUNT(t.id) >= 400 THEN '...' 100 row(s) returned 0.094 sec / 0.000 sec

Ejercicio 4

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades:

```

350 • DELETE from transaction
351 WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
352

```

66 12:59:44 DELETE from transaction WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD' 1 row(s) affected 0.015 sec

Ejercicio 5

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra:

```

359 • CREATE VIEW VistaMarketing AS
360 SELECT
361     c.company_name,
362     c.phone,
363     c.country,
364     AVG(amount) AS Media_compra
365 FROM company c
366 JOIN
367     transaction t ON c.company_id = t.business_id
368 GROUP BY c.company_id, company_name, c.phone, c.country;
369
370 • SELECT * FROM VistaMarketing
371 ORDER BY Media_compra DESC;

```

	company_name	phone	country	Media_compra
►	Ac Fermentum Incorporated	+64 601 244 6320	Germany	330.769338
	Ut Semper Foundation	+10 283 500 6449	Sweden	307.372945
	Orci Adipiscing Limited	+60 897 390 5820	United Kingdom	306.265648
	Dolor Vitae Limited	+6 421 567 9195	France	305.952745
	Nunc In Foundation	+13 267 741 2619	Italy	305.396141
	Maecenas Malesuada Fringilla Inc.	+51 673 540 5289	Netherlands	305.149731
	Aliquet Sem Limited	+44 744 425 9915	Netherlands	304.211717
	Nulla Integer Vulputate Corp.	+75 236 687 9162	Sweden	303.424879
	Non Justo Corp.	+67 963 294 1666	Sweden	303.304574
	Turpis Company	+17 624 311 7168	Netherlands	302.770034
	Aliquam Erat Volutpat LLP	+39 372 330 7229	Italy	301.969666

73 13:24:08 SELECT * FROM VistaMarketing ORDER BY Media_compra DESC

100 row(s) returned

0.344 sec / 0.000 sec

Nivell 3

Ejercicio 1

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon:

Quantes targetes estan actives?

```

379 • CREATE TABLE estat_targetes AS
380 WITH UltimesTransaccions AS (
381     SELECT
382         card_id,
383         declined,
384         ROW_NUMBER() OVER(PARTITION BY card_id ORDER BY timestamp DESC) AS ordre_transaccio
385     FROM transaction
386 )
387 SELECT
388     card_id,
389     CASE
390         WHEN SUM(declined) = 3 THEN 'inactiu'
391         ELSE 'actiu'
392     END AS estat
393 FROM UltimesTransaccions
394 WHERE ordre_transaccio <= 3
395 GROUP BY card_id;

397 • SELECT COUNT(*) AS total_targetes_actives
398 FROM estat_targetes
399 WHERE estat = 'actiu';

```

	total_targetes_actives
▶	5000

		Message	Duration / Fetch
1	10:21:27	CREATE TABLE estat_targetes AS WITH UltimesTransaccions AS (SELECT card_id, declined, ... 5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0	0.360 sec
2	10:21:55	SELECT COUNT(*) AS total_targetes_actives FROM estat_targetes WHERE estat = 'actiu'	1 row(s) returned 0.016 sec / 0.000 sec

Ejercicio 2

Crea una taula amb la qual puguem unir les dades de l'arxiu de products.csv amb la base de dades creada (ja que fins ara no podíem fer-ho), tenint en compte que des de transaction tens product_ids. Genera la següent consulta:

Necessitem conèixer el nombre de vegades que s'ha venut cada producte.

```
403 ● ○ CREATE TABLE product (  
404     id INT PRIMARY KEY,  
405     product_name VARCHAR(100),  
406     price DECIMAL(10, 2),  
407     colour VARCHAR(50),  
408     weight DECIMAL(10, 2),  
409     warehouse_id VARCHAR(50),  
410     category VARCHAR(50),  
411     brand VARCHAR(100),  
412     cost DECIMAL(10, 2),  
413     launch_date DATE  
414 );
```

```

416 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__products.csv'
417 INTO TABLE product
418 FIELDS TERMINATED BY ','
419 ENCLOSED BY '"'
420 LINES TERMINATED BY '\n'
421 IGNORE 1 ROWS
422 (id, product_name, @price, colour, weight, warehouse_id, category, brand, @cost, launch_date)
423 SET
424     price = REPLACE(@price, '$', ''),
425     cost = REPLACE(@cost, '$', '');

```

```

424 • CREATE TABLE transaction_product (
425     transaction_id CHAR(36) NOT NULL,
426     product_id INT NOT NULL,
427     PRIMARY KEY (transaction_id, product_id),
428     FOREIGN KEY (transaction_id) REFERENCES transaction (id),
429     FOREIGN KEY (product_id) REFERENCES product(id)
430 );

```

```

432 • INSERT INTO transaction_product (transaction_id, product_id)
433 SELECT
434     t.id AS transaction_id,
435     jt.product_id
436 FROM transaction t
437 CROSS JOIN JSON_TABLE(
438     CONCAT('[', REPLACE(t.product_ids, ' ', ''), ']'),
439     '$[*]' COLUMNS (
440         product_id INT PATH '$'
441     )
442 ) AS jt;

```

```

445 • SELECT product_id AS Id,
446     (SELECT product_name
447     FROM product
448     WHERE product.id = transaction_product.product_id) AS Producto,
449     COUNT(transaction_id) AS Venta
450 FROM transaction_product
451 WHERE transaction_id = (SELECT id
452     FROM transaction
453     WHERE transaction.id = transaction_product.transaction_id
454     AND declined = 0)
455 GROUP BY product_id
456 ORDER BY Venta DESC;

```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
	Id	Producto	Venta					
	71	Tully Dorne	2437					
	22	chewbacca mustafar	2428					
	19	dooku solo	2423					
	42	sith	2404					
	43	duel	2391					

43 12:56:32 INSERT INTO transaction_product (transaction_id, product_id) SELECT t.id AS transaction_id, p.product_id ... 253388 row(s) affected Records: 253388 Duplicates: 0 Warnings: 0 6.015 sec

44 12:57:04 SELECT product_id AS Id, (SELECT product_name FROM product WHERE product.id = transaction_pr... 100 row(s) returned 0.891 sec / 0.000 sec